THE LEGACY DATA ARCHIVES: APOLLO AND VIKING SEISMIC EXPERIMENTS

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Abstract

About four decades have passed since seismic experiments were performed on the Moon and Mars during the Apollo and Viking missions . These experiments were the huge accomplishments, advancing our understandings in extraterrestrial seismology. With DARTS, these data have now been released to the public in an easily usable format through the Internet. In this report, we show the difficulties we faced with the legacy data archives.

BACKGROUND

A planetary exploration gives us a precious opportunity, and the data provided through such opportunities are also valuable as well. The long-term preservation of the data plays an important role in ensuring significant scientific results. Although archiving activities of on-going missions are important, recovering the legacy data from old missions also enhances the worth of the missions.
About forty years ago, the Apollo and Viking missions began acquisition of important extraterrestrial seismic data. In both missions, the data acquired by seismometers were transmitted to Earth and recorded on reel-to-reel tapes. DARTS [2] at ISAS/JAXA now provides both the Apollo and Viking seismic data with all the available key information. Data restoration and archiving activities were carried out on tapes stored at ISAS/JAXA (Figure 1). These tapes were obtained from the University of Texas at Austin nearly twenty-five years ago as a result of a collaborative effort to migrate the data from 7- and 9-track tapes to 8-mm video cassette tapes [5]. Our legacy data archiving activity began with these 8-mm tapes.

Figure 1: **8-mm video cassette tapes stored at ISAS/JAXA.**

APOLLO SEISMIC EXPERIMENT DATA ARCHIVES

Data coverage:

The archiving of the Apollo data covers all the Passive Seismic Experiment (PSE) data from the Apollo 11, 12, 14, 15 and 16 landing sites. It also includes Apollo 17 experiments: ‘Passive Listening Mode’ data of

LSPE (Lunar Surface Profiling Experiment) and LSG (Lunar Surface Gravimeter). Table 1 shows the data coverage in each experiment.

Apollo Station	Experiments	Start Date (Year/DOY)	Stop Date (Year/DOY)
11	PSE	1969/202 (July 21) 1969/231 (August 19)	1969/215 (August 3) 1969/238 (August 26)
12	PSE	1969/323 (November 19)	1977/273 (September 30)
14	PSE	1971/036 (February 5)	1977/273 (September 30)
15	PSE	1971/212(7/31)	1977/273 (September 30)
16	PSE	1972/112 (April 21)	1977/273 (September 30)
17	LSPE	1976/228 (August 15)	1977/115 (April 25)
17	LSG	1976/061 (March 1) 1977/115 (April 25)	1976/227 (August 14) 1977/273 (September 30)

Table 1: Data coverage of Apollo Passive Seismic Observation Data. **Apollo 14 has some gaps due to transmission losses from the Moon[6].**

Archiving concept:

Our data archiving activity is for the purpose of a long-term preservation of the data to support scientific endevors. Unfortunately, it often requires knowing many details to extract seismic data from the raw data files. Therefore, we started with considering how to reduce such drudgery. There were four significant aspects: accessibility, format, data correction, and document. Also, DARTS provides some tools to support a better understanding of the data.

Accessibility:

The Apollo seismic data can be downloaded from Incorporated Research Institutions for Seismology (IRIS). However, some files are currently missing from the IRIS website. For the tapes stored at ISAS/JAXA, some of them could not be read due to aging degradation, and certain human-induced errors have been corrected later in a small number of files now at UTIG. To make the archives complete, we checked the whole datasets at IRIS, ISAS/JAXA, and UTIG, comparing them with Technical Report 118 [5]. As a result of this comparison, we decided to provide the UTIG's dataset in DARTS.

Format:

UTIG Technical Report No.118 describes the format. Most seismologists use seismic specific formats such as SEED, SAC, etc nowadays, and it is difficult for them to extract seismic data from raw binary files, which involves bitwise operations. Also, legacy raw data such as the Apollo data often contain bit errors. In addition, human errors may have been introduced in the process of data transduction. An example of the latter is a set of 15 files containing data acquired in January 1976 but giving year 1975 in their headers. To inform the user about such errors, DARTS provides a data list with appropriate comments.

Data correction:

For the Apollo data, one data frame consists of 64 ALSEP data words, and most, but not all, even-numbered ALSEP words are assigned to the SPZ component of the PSE data with a few even-numberd words allocated to other experiments. Thus, to achieve an evenly spaced time series for the SPZ component, we must interpolate the skipped data values. For this correction, a 4th degree Lagrange interpolation is applied.

Document:

There are multiple documents related to the Apollo seismic experiments and necessary information needed to accomplish a given task may be spread out in several documents. It often is a difficult task to read all these documents and search for required information in them. To help the user in this respect, two documents have been prepared for DARTS: One is the general description and the operational status of the Apollo Seismic Experiments, and the other is the frequency responses of the Apollo seismometers.

Database system:

A relational database is utilized to manage the extracted data. The whole data are stored in the database once, and one can access the data using SQL query (Figure 2). Stored functions on the database are also implemented to create time series data (Figure 3).

```
apollo=> SELECT time,lp_x,lp_y,lp_z FROM tbl_pse WHERE file_id = '139' LIMIT 5;
          time           |       lp_x        |       lp_y        |       lp_z
-------------------------+-------------------+-------------------+-------------------
 1970-03-13 14:21:21.18  | {524,524,524,524} | {494,494,494,494} | {497,497,496,496}
 1970-03-13 14:21:21.784 | {524,524,524,524} | {494,494,494,494} | {496,496,496,496}
 1970-03-13 14:21:22.388 | {524,524,524,524} | {494,494,494,494} | {496,496,496,496}
 1970-03-13 14:21:22.992 | {524,524,524,524} | {494,494,494,495} | {496,496,496,496}
 1970-03-13 14:21:23.595 | {524,524,524,523} | {495,495,495,495} | {496,497,496,496}
(5 rows)
```

Figure 2: **Apollo seismic data on PostgreSQL.**

```
apollo=> SELECT ap_station,frame_count,nc,time,lp_x,lp_y,lp_z FROM alsep_lp('1970-03-1
NOTICE:  stmt= SELECT * FROM (SELECT e.ap_station, e.ground_station,  e.file_id, e.pos
al)::timestamp without time zone as time, e.nc, e.lp_x, e.lp_y, e.lp_z,  e.process_fla
s, f.length, f.frame_count,  f.time_diff, f.time, generate_series(1, 4) nc, unnest(f.l
flag FROM tbl_pse f WHERE '1970-03-13 14:21:20.396' <= f.time AND f.time <= '1970-03-1
ime AND d.time < '1970-03-13 14:21:22.51'
CONTEXT:  PL/pgSQL function "alsep_lp" line 26 at assignment
 ap_station | frame_count | nc |          time           | lp_x | lp_y | lp_z
------------+-------------+----+-------------------------+------+------+------
         12 |          87 |  1 | 1970-03-13 14:21:21.18  |  524 |  494 |  497
         12 |          87 |  2 | 1970-03-13 14:21:21.331 |  524 |  494 |  497
         12 |          87 |  3 | 1970-03-13 14:21:21.482 |  524 |  494 |  496
         12 |          87 |  4 | 1970-03-13 14:21:21.633 |  524 |  494 |  496
         12 |          88 |  1 | 1970-03-13 14:21:21.784 |  524 |  494 |  496
         12 |          88 |  2 | 1970-03-13 14:21:21.935 |  524 |  494 |  496
         12 |          88 |  3 | 1970-03-13 14:21:22.086 |  524 |  494 |  496
         12 |          88 |  4 | 1970-03-13 14:21:22.237 |  524 |  494 |  496
         12 |          89 |  1 | 1970-03-13 14:21:22.388 |  524 |  494 |  496
```

Figure 3: **An access example using stored function: alsep_lp.**

Tools:

We have prepared a browser-based data viewer: ALSEP Data Viewer (Figure 4). This tool graphically shows the ALSEP word allocation of the decoded data.

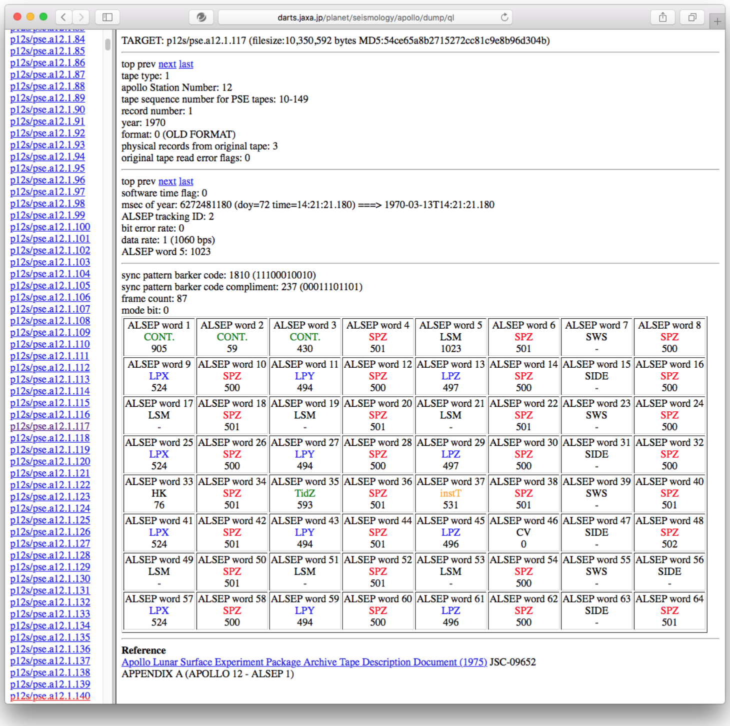

Figure 4: **A browser-based data viewer: ALSEP Data Viewer.**

An open source console program, alsep_tool, is also provided from DARTS (Figure 5). It decodes the raw data file and dumps data in ASCII. This tool also includes a program to build a database. To show the seismic waveform of a specific event, the Moon Seismic Monitor is developed (Figure 6).

```
[$ pseinfo -rfd pse.a12.1.117
--- record info ---
offset in file: 0
tape type: 1
apollo station: 12
tape seq: 10149
record number: 1
year: 1970
format: 0
phys records: 3
read error: 0
error flag: 0

frame info: frame_count=87,doy=72,time=14:21:21.180,msec=6272481180,diff=6272481180,track_id=2,sync_code=1810
#spz: 501,501,501,500,500,500,500,500,501,501,501,500,500,500,500,500,501,501,501,501,501,501,502,502,501,501
```

Figure 5: **Open source program alsep_tools is prepared to extract data in ASCII from original Apollo files.**

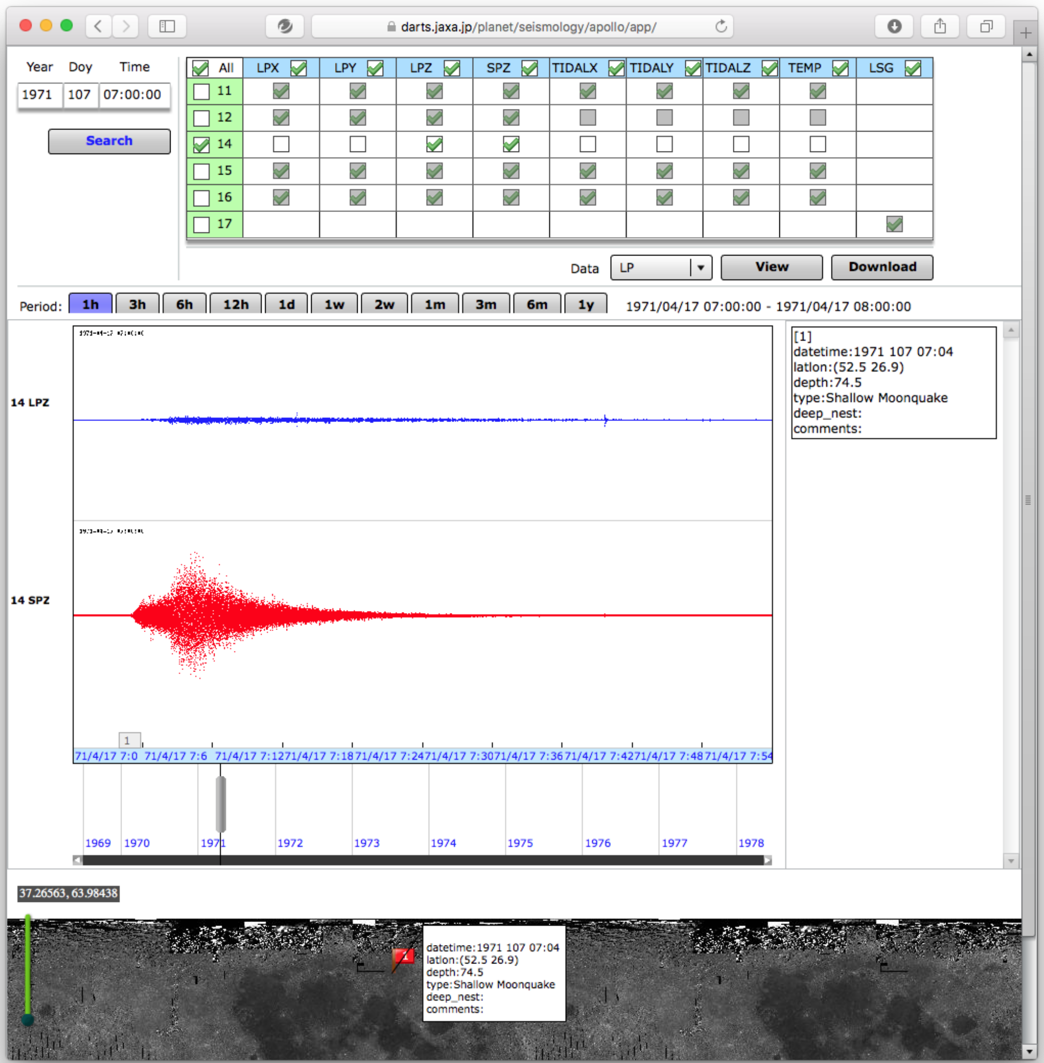

Figure 6: **A browser-based application Moon Seismic Monitor.**

VIKING LANDER2 SEISMIC EXPERIMENT DATA ARCHIVES

Difference from Apollo archives:

The seismic data of the Viking Lander 2 were included in the Apollo 8-mm tapes [5]. After the Apollo archives had been worked to some extent, we started to investigate the Viking tapes. This effort, however, was harder than we anticipated. The data recovery from the seismometer on Viking Lander 2 differed from that of the Apollo archives. In contrast to the highly successful Apollo seismic experiments, the Viking Lander 2 mission was unable to fully reveal the seismic activities on Mars. Due to this or for some other reasons, the information needed to build data archives was extremely scanty. Hence, we started with collecting whatever available information. Any small hints, such as 6-bit dumped data with pencilled-in notes or assembler source codes of an old PDP-15 computer, were helpful.

VUS Tape:

Technical Report 118 [5] includes descriptions of two types of Viking seismic data files: SEIS and USEIS. The former contains raw data streams as received from Mars and recorded on SEISF tapes, while the latter contains unwrapped seismic data as recorded on VUS tapes. Analysis of the VUS tape files at ISAS/JAXA started from handwritten notes found at UTIG (Figure 7).

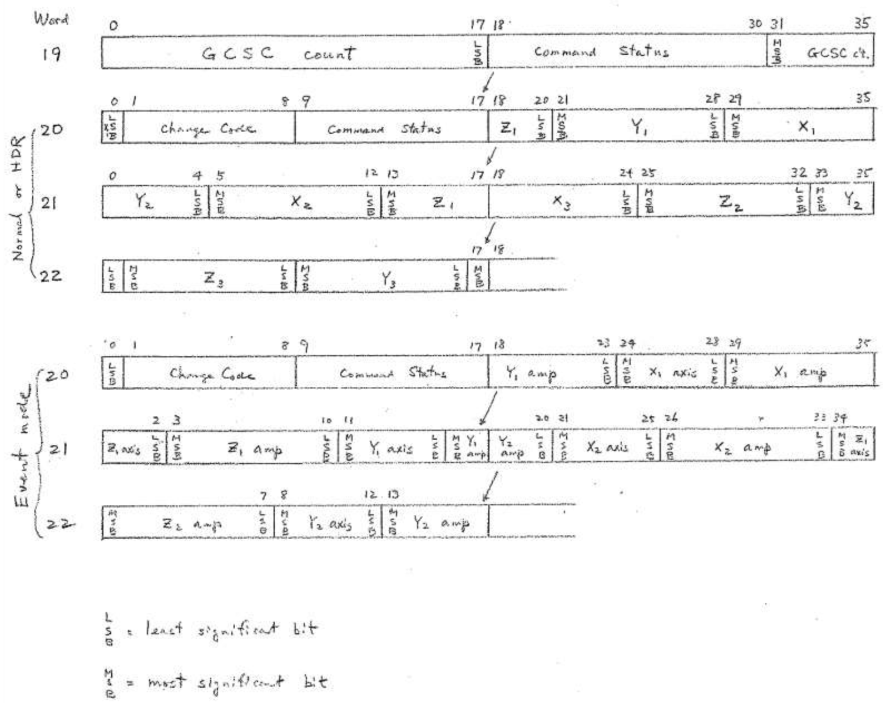

Figure 7: Handwritten note of VUS tape format found at UTIG. **Format analysis at ISAS/JAXA started with this note.**

EDR-2 Tape:

While analysing the VUS format files, we found another set of Viking seismic data at the NSSDC website. The data bear NSSDC ID: PSPG-00472 and PSPG-00070. We obtained these data from NSSDC and investigated their formats. Although PSPG-00070 is an ASCII file (Figure 8), it was hard to understand. Also, the content of PSPG-00472 looked like binary data at a first glance. There were many H'40 in PSPG-00472, which enabled us to notice that it was a padding, and the code was in EBCDIC. With this hint, the analysis went smoothly. PSPG-00472 contains a PL/I source code to convert the data format from UNIVAC 1108 to IBM 370. In this source code, the keyword EDR was found. Using the keyword EDR in a search engine of the Internet, we found the EDR-2 document [7] that described the detailed format. However, there were some unprintable characters even in the decoded ASCII file. It was the separator of NSSDC's Archival Information Package (AIP) [3]. This information was described as an attribute file attached to PSPG-00070. Thus, we were able to restore the EDR-2 tape files completely (Figure 9).

```
 0    0 0    0 0    0 0 GGC N    0 0    0 0    0 0    0 0    0 0    0 0    0 0    0 0    0 0    0 0    0 0    0 0 FGC    2    1050     0143072
47  44925593   0833HIGH    0DB   18DB    X20   2.HZ FFC [ STEP   XOFF   YON    ZOFF   CALOFF 1976       36922      .75476540+05       HFC
        3 0 106 0    0 0   42 0-112 0    1 0   61 0    0 0   37 0 -56 0-128 0   24 0    GHC [-128 0    2 0 102 0   64 0    5 0   64 0  97
 0   50 0 -96 0 -58 0-104 0   44 0 GGC ]-119 0   77 0 -69 0    2 0    9 0-128 0    3 0   37 0    0 0    5 0 -64 0-128 0 GGC     10 0-1
26 0    0 0   32 0   72 0    9 0   64 0    1 0   16 0 -96 0    6 0 -96 0 GGC   -128 0   56 0   96 0   87 0 -48 0-128 0   78 0 -88 0    1 0
   0 0   62 0   32 0 GGC   -121 0 -96 0-128 0   27 0-108 0    1 0   80 0   64 0   13 0 -55 0    9 0   47 0 GGC A -32 0    2 0 100 0-128
 0   30 0 112 0    1 0F B -14 0   64 0   11 0 -62 0    1 0 GGC B   57 0   34 0 -96 0    0 0   25 0   16 0 -95 0   80 0-128 0   12 0 -84
0    0 0 GGC C -88 0   64 0    6 0   84 0    0 0   20 0   32 0    6 0 106 0    0 0   10 0-103 0 GGC D    5 0   54 0   32 0    5 0 -16 0-12
8 0   63 0 -96 0    2 0 108 0    0 0 -97 0 GGC E   92 0 -39 0    1 0   57 0 -64 0   12 0 -28 0-128 0   12 0 -30 0    2 0 115 0 GGC F
 64 0    4 0 -80 0   65 0-103 0   32 0    7 0   56 0-126 0   28 0-112 0-125 0 GGC G   92 0   68 0   14 0 -55 0    1 0 -82 0    5 0 -32
0-128 0 -67 0 -48 0    2 0 GGC H 104 0   64 0   14 0 104 0   34 0   24 0 113 0    7 0   20 0   32 0   92 0-128 0 GGC I    3 0   92 0
0 0-114 0 -56 0    1 0   46 0    0 0    1 0 -27 0    1 0    7 0 GGC J-126 0    1 0 114 0-128 0   11 0   40 0 -56 0    1 0   14 0   32 0
 22 0 102 0 GGC K    0 0   23 0-128 0    3 0   54 0-128 0   43 0 -56 0    1 0   59 0   64 0    5 0 GGC L -32 0-128 0   12 0 -96 0    2
0 114 0   64 0    2 0 112 0    1 0   32 0   23 0 GGC M -28 0    0 0   23 0-112 0    1 0   50 0-128 0    9 0-104 0   33 0   57 0 -48 0 GG
C N    5 0 -28 0-128 0   12 0 -32 0   18 0   48 0   64 0    6 0 112 0-127 0 121 0 DGC O   32 0-109 0   56 0-128 0   28 0-112 0    3 0
 92 0   64 0 FDC    2    1060      013992247   44696334   0512EVENT 18DB    6DB    X12   2.HZ FFC [ FIX    XON    YOFF   ZOFF   CAL ON 197
```

Figure 8: Raw data example of EDR-2 Tape (PSPG-00070).

```
FGC    2    1050     014307247   44925593   0833HIGH    0DB   18DB    X20   2.HZ
 FFC [ STEP   XOFF   YON    ZOFF   CALOFF 1976       36922      .75476540+05
 HFC           3 0 106 0    0 0   42 0-112 0    1 0   61 0    0 0   37 0 -56 0-128 0   24 0
 GHC [-128 0    2 0 102 0   64 0    5 0   64 0   97 0   50 0 -96 0 -58 0-104 0   44 0
 GGC ]-119 0   77 0 -69 0    2 0    9 0-128 0    3 0   37 0    0 0    5 0 -64 0-128 0
 GGC     10 0-126 0    0 0   32 0   72 0    9 0   64 0    1 0   16 0 -96 0    6 0 -96 0
 GGC   -128 0   56 0   96 0   87 0 -48 0-128 0   78 0 -88 0    1 0    0 0   62 0   32 0
 GGC   -121 0 -96 0-128 0   27 0-108 0    1 0   80 0   64 0   13 0 -55 0    9 0   47 0
 GGC A -32 0    2 0 100 0-128 0   30 0 112 0    1 0 -14 0   64 0   11 0 -62 0    1 0
 GGC B  57 0   34 0 -96 0    0 0   25 0   16 0 -95 0   80 0-128 0   12 0 -84 0    0 0
 GGC C -88 0   64 0    6 0   84 0    0 0   20 0   32 0    6 0 106 0    0 0   10 0-103 0
 GGC D   5 0   54 0   32 0    5 0 -16 0-128 0   63 0 -96 0    2 0 108 0    0 0 -97 0
 GGC E  92 0 -39 0    1 0   57 0 -64 0   12 0 -28 0-128 0   12 0 -30 0    2 0 115 0
 GGC F  64 0    4 0 -80 0   65 0-103 0   32 0    7 0   56 0-126 0   28 0-112 0-125 0
 GGC G  92 0   68 0   14 0 -55 0    1 0 -82 0    5 0 -32 0-128 0 -67 0 -48 0    2 0
 GGC H 104 0   64 0   14 0 104 0   34 0   24 0 113 0    7 0   20 0   32 0   92 0-128 0
 GGC I   3 0   92 0    0 0-114 0 -56 0    1 0   46 0    0 0    1 0 -27 0    1 0    7 0
 GGC J-126 0    1 0 114 0-128 0   11 0   40 0 -56 0    1 0   14 0   32 0   22 0 102 0
 GGC K   0 0   23 0-128 0    3 0   54 0-128 0   43 0 -56 0    1 0   59 0   64 0    5 0
 GGC L -32 0-128 0   12 0 -96 0    2 0 114 0   64 0    2 0 112 0    1 0   32 0   23 0
 GGC M -28 0    0 0   23 0-112 0    1 0   50 0-128 0    9 0-104 0   33 0   57 0 -48 0
 GGC N   5 0 -28 0-128 0   12 0 -32 0   18 0   48 0   64 0    6 0 112 0-127 0 121 0
 DGC O  32 0-109 0   56 0-128 0   28 0-112 0    3 0   92 0   64 0
```

***Figure 9: Preformat data example of EDR-2 Tape (PSPG-00070)*. Red, blue, and green corresponds to X, Y and Z, respectively.**

Time System:

Before analysing the Viking seismic data, the time system used in Viking must be learned.

(1) SOL: SOL is the length of a Martian solar day, which is 88,775.244 seconds (24h 39m 35.244s). On Viking 2 Lander mission, SOL is used to express the elapsed time from the local midnight immediately prior to touchdown which was September 3, 1976, 12:48:45 UTC in Earth time [4].
(2) GCSC Count: Guidance Command and Sequencing Computer (GCSC) Count was used for timing management on the lander. One bit represents 160 milliseconds, and GCSC count cycles through 0 approximately every 31 days.
(3) LLT: Local Lander Time (LLT) is the time at the location of the landing point. The unit of LLT is SOL, and LLT ran from 00:00:00 to 24:39:35.244, where hours, minutes, and seconds are equivalent to those of Earth. 00:00:00 LLT corresponds to midnight and 12:19:47.622 corresponds to noon.
(4) Earth-Received Time: The EDR-2 tapes include GMT in each data buffer, which is the time in milliseconds of earth receipt of the first bit of the buffer. This time is used as a guide to convert GCSC Count to LLT.

It is often required to convert GCSC count to LLT because most of the published reports express time in LLT while the acquired data are referenced to GCSC count. The parameters to be used for this conversion are subject to periodic calibrations, and will be discussed elsewhere later.

Verification:

At first, we checked the difference between the VUS-tape and EDR-2-tape files. Most of the data are essentially identical. However, some errors are found both in VUS and EDR-2 files. Therefore, we need to continue the investigation to create a best possible dataset.
Secondary, we investigated the EDR-2 files against the figures on the report: 'Seismology on Mars' [1]. This comparison was needed to verify whether our restoration was correct. Figure 10 shows an example of such a verification effort. Our restoration approach proved to be successful.

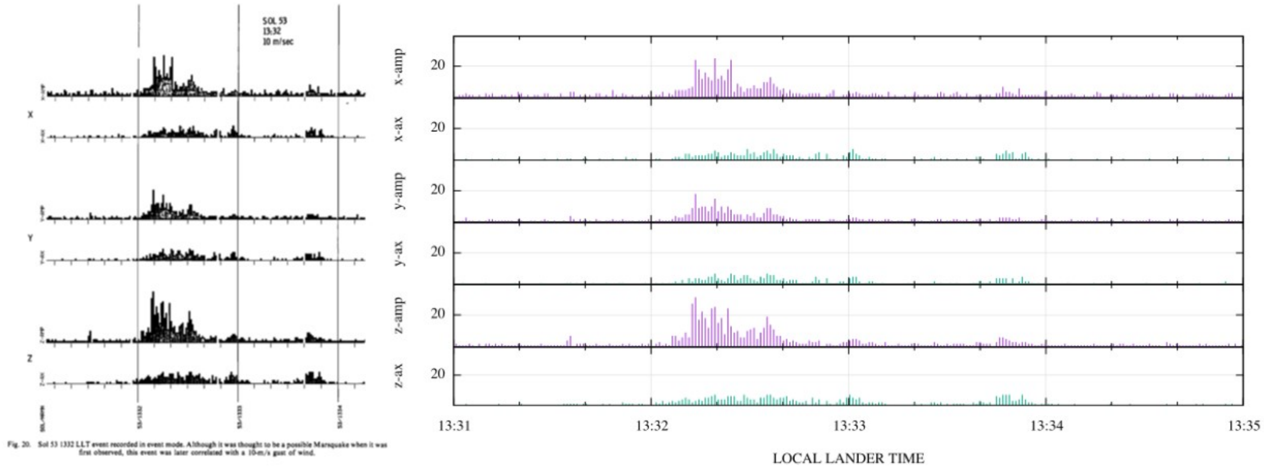

Figure 10: **A verification example of restored Viking seismic data. On the left is figure 20 of the report: Seismology on Mars [1], and on the right is reconstructed data from an EDR-2-tape file.**

CONCLUSION

We performed the legacy data archives for the Apollo seismic experiments and Viking Lander 2 mission. There is a significant difference between the problems each data set posed. For the Apollo missions, integrating dispersed information was the major task. For the Viking Lander 2 mission, it was necessary to utilize small hints reflecting from our experiences with computers. Therefore, the Viking case was like archaeology, and data archivists must remember that legacy data are not always restored.

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